

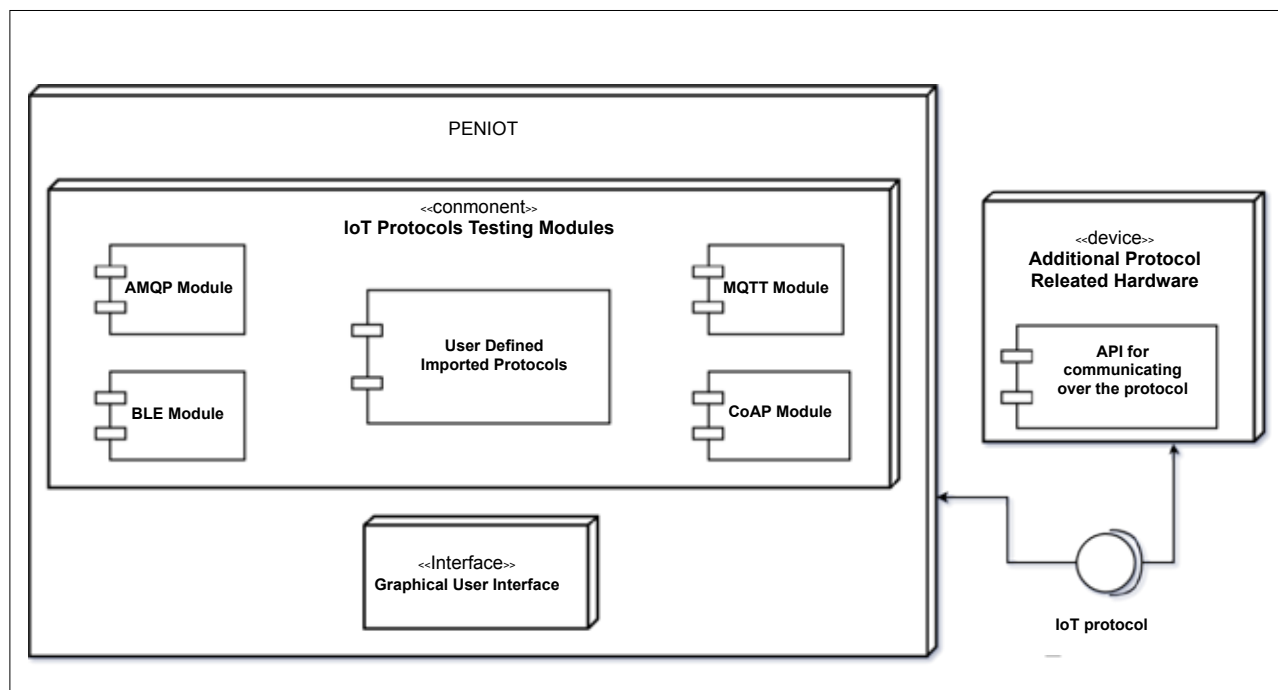


Figure 1: Key components in PENIOT

The need for security audit in IoT gadgets

Cyber attacks are evolving all the time, and businesses and government agencies are employing ever more sophisticated cyber security measures to guard their applications and infrastructure from online attacks. The global penetration testing market is expected to grow from US\$ 1.6 billion in 2021 to US\$ 3.0 billion by 2026, at a CAGR of 13.8 per cent between 2021 and 2026.

Penetration testing on IoT devices is a hot topic and there is huge research in this segment. Even with a 'security by design' approach, pen testing is vital to identify true security dangers and take the appropriate precautions.

Key segments and protocols in IoT deployment where security and privacy is required are:

- Constraint application protocol (CoAP)
- Bluetooth low energy (BLE)
- Advanced message queuing



Figure 2: The software suite of Objection for smart gadgets

protocol (AMQP)

- Message queuing telemetry transport (MQTT)
- For an attacker to get access to a connected device, there are several possible entry points. At the time of an IoT pen test (or security audit), the complete IoT scenario and ecosystem is tested. This encompasses everything from the individual layer and the embedded software, to the communication protocols and

the server. Tests of servers, online interfaces, and mobile apps are not unique to IoT, but are crucial since they cover areas with a high potential for failure. Vulnerabilities in IoT are the focus of electrical, embedded software, and communication protocol testing.

The following tests are run during a connected device's security assessment. These are done using different high performance penetration testing and security audit tools